

Rotational Motion

$$\text{Torque} = I \times \alpha$$

$$\text{Angular momentum } L = I \times \omega$$

$$\text{Ring: } I = MR^2 \quad \text{Disc: } I = MR^2 / 2$$

If external torque = 0, L is conserved

Laws of Motion

$$\mathbf{F} = m\mathbf{a}$$

$$\text{Friction: } f \leq \mu N$$

Impulse = change in momentum

Free-body diagram first, always!

Chemical Bonding

Ionic: transfer of electrons

Covalent: sharing of electrons

Bond order = $(N_b - N_a) / 2$

Higher bond order = shorter bond

Cell: The Unit of Life

Prokaryotes: no nuclear membrane

Mitochondria: powerhouse (ATP)

Ribosomes: protein synthesis

70S in prokaryotes, 80S in eukaryotes

Practice Questions (already white)

1. A disc of mass 2 kg and radius 0.5 m rotates at 10 rad/s. Find L.
2. State the law of conservation of angular momentum.
3. Why does an ice skater spin faster with arms folded?